

Bitcoin Volatility Prediction

A Time-Series Forecasting Analysis

73,000+ Hourly BTC/USDT Records

Introduction

Why turbulence, not direction?

Bitcoin's price swings unpredictably at short, hourly timescales, and traders, exchanges, and risk desks have no reliable way to anticipate how turbulent the next stretch of trading will be. That leads to mispriced risk, thin margin buffers, and poorly-timed position sizing.

AN ANALOGY

Forecasting price is like predicting exactly where a boat will be tomorrow. Forecasting volatility is like predicting how rough the sea will be. The second question is more tractable - and for anyone managing risk, it's usually the more useful one.

Why Volatility, Not Price

Use Case	How Volatility Enters
Option pricing	Black-Scholes takes sigma as a required input
Value-at-Risk	Loss estimates scale directly with sigma
Position sizing	Trade smaller when choppy, larger when calm
Exchange margin	Collateral requirements rise with volatility

Problem Statement

From a vague question to a testable target

THE AIM

Predict Bitcoin's 7-day realized volatility - the rolling standard deviation of hourly returns over a 168-hour window - 168 hours ahead of the features used to predict it. This gives traders and risk desks a continuous, hour-by-hour-updating estimate of next week's volatility, usable directly in position-sizing and margin decisions.

Target Definition

Target	volatility_7d, 168h ahead of the features
Horizon	168 hours (7 days) - never the next hour
Features	Lagged price/volume history + technical indicators
Success bar	Beat naive persistence and Holt-Winters baselines

PROBLEM TYPE: REGRESSION

Downstream consumers of a volatility number - Black-Scholes sigma, Value-at-Risk, position-sizing formulas - need an actual continuous estimate, not a high/low category. That rules out classification from the outset.

Objective: 7-Day Volatility Forecasting

GOAL

Forecast Bitcoin's 7-day realized volatility from hourly price data. The key objective is not to claim a perfect crystal ball, but to show how EDA and feature engineering set realistic expectations, justify modeling choices, and clearly explain what this workflow can and cannot deliver.

Workflow Stages

- 1. EDA Insights:** What patterns does volatility actually follow?
- 2. Feature Engineering:** Which signals carry real, non-redundant information?
- 3. Modeling & Results:** Can that signal be turned into a working forecast?
- 4. Limitations & Recommendations:** What would genuinely move the needle further?

Data Overview & Feature Dictionary

DATASET: 73,000+ hourly BTC/USDT records

FEATURES: Price, volume, and engineered volatility regressors

TARGET: 7-day realized volatility, 168 hours ahead

Feature	Description	Type
RV_day / RV_week / RV_month	Volatility over the past day / week / month (HAR horizons)	Numeric
ATR_X	Average true range, as a % of price	Numeric
EWMA_X	EWMA-smoothed volatility ($\lambda = 0.995$)	Numeric
SKEW_X	Realized skewness of hourly returns	Numeric
hourly_return	Hour-over-hour price return	Numeric
rsi_14 / macd_hist / bb_width	Momentum and dispersion indicators	Numeric
volatility_7d (target)	7-day realized volatility, 168h ahead of the features	Target

Dataset & Data Cleaning

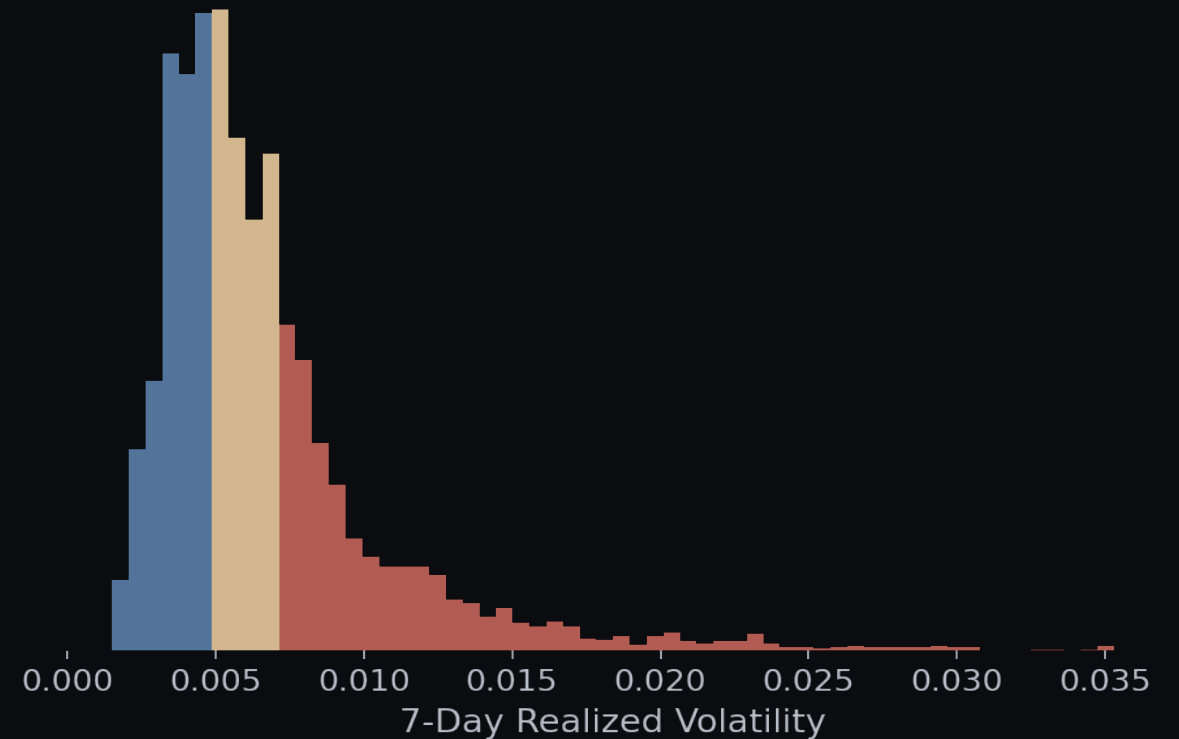
Data Cleaning Steps

1 Removed non-trading gaps and invalid candles (28 timestamp gaps identified)

2 Guarded zero-volume candles before rolling calculations

3 Standardized hourly timestamp frequency across the full history

4 Dropped rolling-window warm-up rows (first ~30 days of history)



Regime Mapping

Level	Regime	Tertile
0	Low	≤ 0.0047
1	Moderate	0.0047 - 0.0070
2	High	> 0.0070

| Feature Engineering

01

HAR Horizons

Aggregated volatility at day / week / month windows to capture the long-memory pattern realized-volatility literature expects (Corsi, 2009).

02

ATR as % of Price

Fixed a real bug: raw dollar-terms ATR was a ~45,000x scale mismatch against every other regressor. Fixing it was the single biggest gain in the project.

03

EWMA Smoothing

Backtested the industry-standard exponential decay factor ($\lambda = 0.995$) and added it as its own regressor, not just a separate model.

04

Realized Skewness

Captures return-distribution shape, not just magnitude - a genuinely different signal from the volatility-level regressors above.

EDA: Temporal Features

Does when something happens carry any signal about how severe it will be?

Autocorrelation (Primary Signal)

ACF DECAYS SLOWLY

The target's autocorrelation function decays slowly across many lags rather than dropping to zero quickly - confirmation that volatility clustering is real and exploitable, not noise. This is the signal HAR-RV-X's three time horizons are built to capture.

Hour-of-Day / Day-of-Week

Test	Result
Chi-square statistic	2.68
Degrees of freedom	46
p-value	1.000000

LIMITED / NO SIGNAL

No significant association between hour-of-day and volatility regime ($p \geq 0.05$). Holt-Winters' own seasonal parameter converges to ~ 0 independently - two different methods agreeing time-of-day isn't real signal here.

TAKEAWAY

Volatility's own recent history carries real signal; the calendar (hour, day-of-week) does not. This is exactly why HAR-RV-X's regressors are all volatility measures, not time-of-day features.

EDA: Range & Dispersion Features

Range-based measures (ATR, Bollinger width) vs. the target.

Two Independent-Looking Measures

ATR and Bollinger width are both built from intraday high/low ranges, not close-to-close returns - a genuinely different signal from simple returns.

Correlation With Target

bb_width correlates at +0.588; atr_14 at +0.142. Both real, but likely overlapping with each other, not two independent sources of signal.

Redundancy, Not Independence

Bollinger width is largely redundant with ATR - both are range-based dispersion measures built from the same underlying price action.

Feature	Correlation with Target
bb_width	+0.588
atr_14	+0.142

Both range measures show dependence with the target, but bb_width's strength is mostly explained by its overlap with atr_14, not by adding new information. This is confirmed directly in Feature Selection (next).

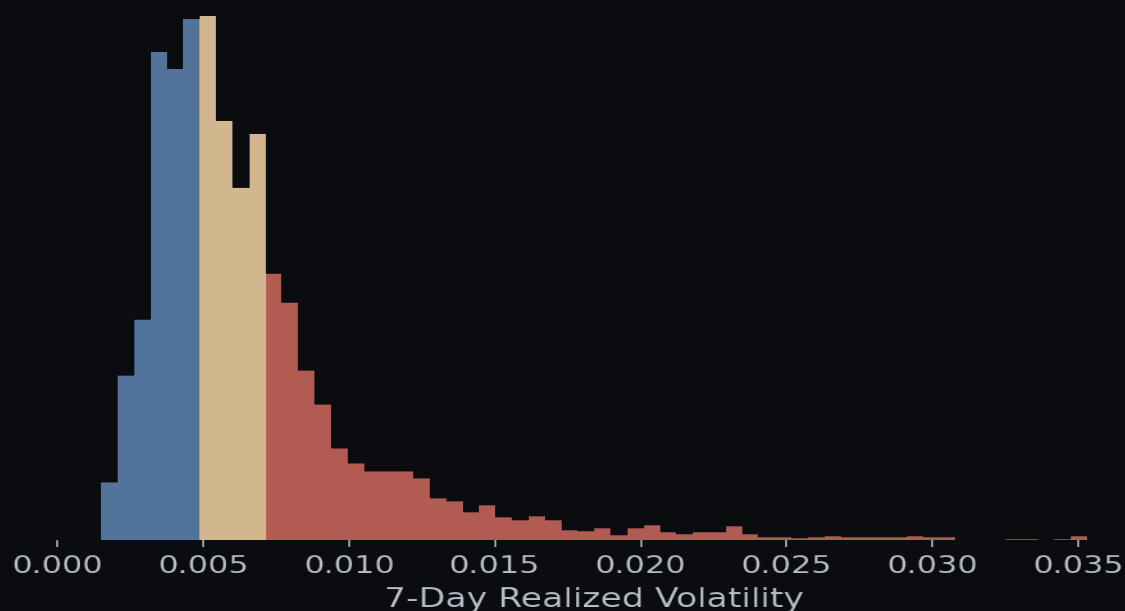
EDA TAKEAWAY

Range-based features carry the strongest raw correlation with the target of anything tested - but that signal is concentrated, not spread across many independent features.

EDA: Volatility Clustering

Do calm and turbulent periods actually separate cleanly?

Certain stretches of history show clearly higher volatility than others, and the target's fat right tail (long calm stretches, sharp spikes) is visible directly in its own distribution (Slide 4). But the boundary between “calm” and “turbulent” is fuzzy, not sharp: regimes cluster in time, but overlap substantially at any given threshold.



TAKEAWAY

Clustering is real and exploitable (HAR-RV-X leans on it directly), but it does not uniquely determine severity - a ceiling on achievable accuracy, not a guarantee of it.

Feature Selection: Correlation & Effect Sizes

Correlation Analysis: all continuous features tested against the target; Chi-Square confirms hour-of-day carries no signal ($p = 1.000$).

Retained (used by HAR-RV-X)

RV_day / RV_week / RV_month

Theoretically motivated (Corsi 2009)

ATR_X

$r = 0.142$, scale-fixed

EWMA_X

backtested smoothing

SKEW_X

distribution shape, not magnitude

Dropped (weak or redundant)

rsi_14

$r = 0.004$

macd_hist

$r = 0.013$

volume_chg

$r = 0.002$

Close / ma_7d (raw levels)

$r = -0.26$, non-stationary, spurious

CRITICAL FINDING

Even the strongest raw correlation (bb_width, +0.588) is largely redundant with ATR, not new information. This is why R^2 tops out around 0.24, not 0.9 - a ceiling set by the data itself, not by model choice.

Feature Relationships: Correlation Analysis

Statistical Measure

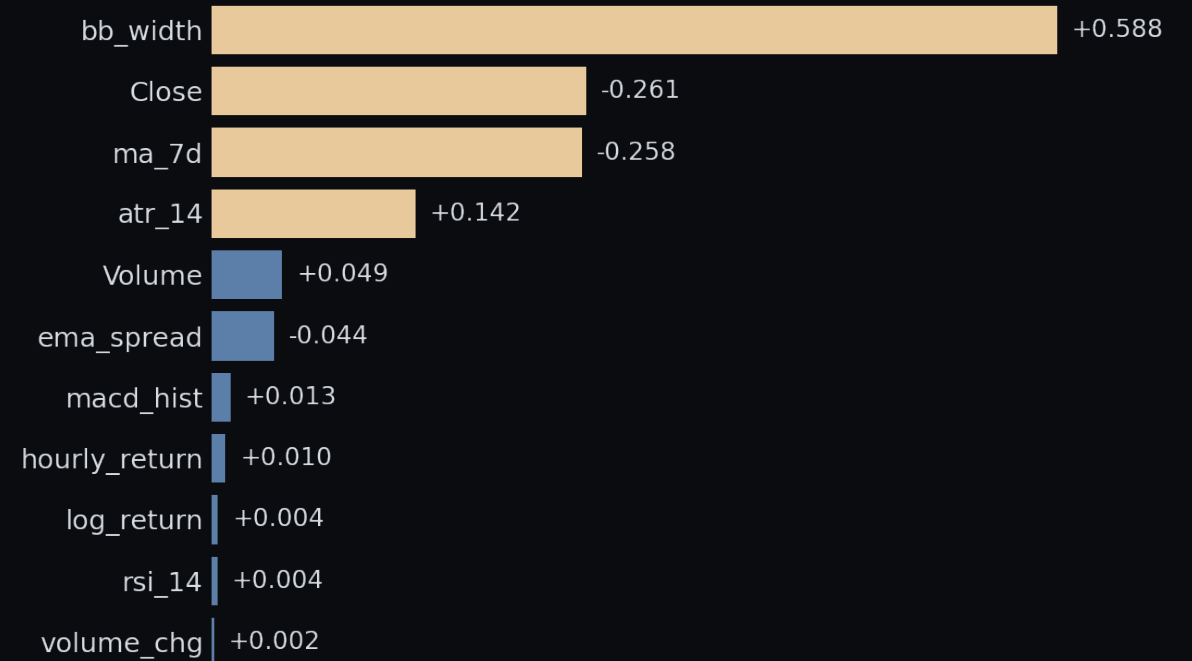
Pearson correlation measures the linear relationship between each continuous feature and the target, on a scale from -1 to +1.

Key Finding

Only range-based features (bb_width, atr_14) show meaningful correlation. Every other continuous feature sits below $|0.05|$ - essentially no linear relationship with the target at all.

Performance Ceiling

This result alone suggested - before any modeling - that performance would be limited. Models cannot exceed the information content of their features.



Model 1: GARCH

GJR-GARCH with Student-t Errors (Tuned)

Model Configuration

Vol Model	GJR-GARCH
Distribution	Student's t
p / o / q	1 / 1 / 1
AIC (Tuned)	96,564.5

Performance (Historical Test)

R2 (Historical)

-5.02

RMSE

0.003876

MAPE

90.54%

KEY INSIGHT

GARCH's forecast mean-reverts toward a long-run average well before 168 hours are up - a structural weakness at this horizon, not an implementation bug. It degrades even further on live data ($R^2 = -110.68$).

Model 2: HAR-RV-X

Regression on Realized-Volatility Horizons, Ridge-Regularized (Final)

Model Configuration

Regressors	6 (RV horizons, ATR_X, EWMA_X, SKEW_X)
Regularization	Ridge
Alpha	0.01
Target Transform	$\log(\text{target})$

Performance (Historical Test)

R² (Historical)

0.242

RMSE

0.001367

MAPE

22.83%

KEY INSIGHT

Best performer among all models, on both historical and genuinely unseen live data ($R^2 = 0.178$ live). A real scale bug fix in ATR_X (a ~45,000x mismatch) was the single largest driver of this result.

Model 3: EWMA

RiskMetrics-Style Exponential Smoothing (Backtested)

Model Configuration

Formula	RiskMetrics (J.P. Morgan, 1996)
Lambda (Backtested)	0.995
Search Range	0.85 - 0.995
Selection	TimeSeriesSplit CV

Performance (Historical Test)

R2 (Historical)

0.059

RMSE

0.001532

MAPE

23.71%

KEY INSIGHT

A single, fully transparent parameter - no hidden layers, nothing opaque. A solid runner-up, and the risk desk's own trusted estimator, but not as sharp as HAR-RV-X's multi-horizon regressors.

Model Comparison

Historical Test-Set Performance Across All Models

Model	RMSE	R2	MAPE	Beats Persistence?
Naive Persistence (baseline)	0.001649	-0.090	24.41%	Baseline
Holt-Winters (baseline)	0.002845	-2.245	43.20%	No
GARCH (GJR, Tuned)	0.003876	-5.023	90.54%	No
HAR-RV-X (Final)	0.001367	0.242	22.83%	Yes
EWMA (lambda=0.995)	0.001532	0.059	23.71%	Yes

KEY FINDING

HAR-RV-X (Final) wins on every metric, beating naive persistence by 17.1% lower RMSE. Both HAR-RV-X and EWMA clear the persistence bar; GARCH and Holt-Winters do not. This gap holds up on genuinely unseen live data too (Section: Live Testing), not just this historical split.

Limitations

Category 01

Data Constraints

- No fundamental or on-chain signals (only price/volume history).
- News, regulation, and macro shocks are invisible to the model.
- Single asset - no cross-crypto spillover signal used.

Category 02

Modeling Constraints

- GARCH's mean-reversion is structural, not tunable away.
- Twenty-two robustness checks tested; most did not beat HAR-RV-X.
- Hardware/time limited exhaustive deep-learning exploration.

Category 03

Statistical Ceiling

- Feature correlations with the target are inherently weak.
- Models cannot exceed the information content of their inputs.
- A modest R2 ceiling is a predictable result of that signal, not a bug.

KEY TAKEAWAY

Meaningful improvement needs genuinely new information (news sentiment, on-chain metrics), not further tuning of what's already here. Neither more model complexity nor more compute alone is sufficient.

Operational Implications

Where a 0.24 R2 forecast is genuinely useful - and where it is not.

Appropriate Uses

Position Sizing

Scale trade size to the forecast - bigger expected move, smaller position.

Options Pricing Input

One data-driven volatility estimate to compare against market pricing.

Risk Reporting

A continuous VaR input for weekly risk reviews, not a single trade call.

Inappropriate Uses

Sole Trading Signal

Explains ~24% of variance - never trade on this number alone.

Real-Time Execution

Not built or tested for split-second, automated trade triggers.

Certainty in Legal/Compliance Filings

A forecast with real error bars, not a guarantee.

One input, not the final answer.

CONCLUSION

Purpose-Built Regressors, Not More Complexity

EDA and feature selection revealed weak, concentrated correlations that set a real ceiling on achievable accuracy. Every attempt to beat that ceiling with a more flexible model - XGBoost, LSTM, ARIMA - lost to HAR-RV-X's three theoretically-motivated regressors.

This is a feature-information problem, not a model-architecture problem.